

20 x 20 Deck · Castle Rock · Vintage PVC · Fulton + drink rail

20' x 20' · 10' high · ledgered · south Castle Rock · 2x10 SYP @ 12" on two 4x12 DF beams, 8 Diamond Piers · TimberTech Vintage Coastline PVC, picture frame, no butt joints · Fulton black + drink rail · Hardie skirt · straight-run stair, 16 risers

CHECK OR ACH	FINANCED	MONTHLY	SQUARE FEET	FOOTINGS	RAIL
\$65,806	\$70,759	\$821	396	12	54 LF

DESIGN READS

Frame	19'-9 7/16" x 19'-9 1/2" outside to outside
Deck	19'-11 3/8" x 19'-10 1/2" over fascia · 396 SF · 10'-0" above grade
Joists	21 2x10 #1 True Frame Joist SYP GC @ 12" OC x 19'-5"
Rims	2-ply 2x10 front and sides
Beam	DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb)
Beam	DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb)
Posts	12 x 6x6 on DP-50/50
Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Vintage Coastline — 41 rows parallel to the house, picture frame in Vintage Coastline, 3/16" gaps, Cortex
Rail	Fulton 36" Black: 8 bays, 10 posts, 54 LF + drink rail
Stair	right: 16 risers @ 7.50", 15 treads, 5 stringers, 4'-0" wide

- nominal 20'-0" x 20'-0" -> frame 19'-9 7/16" x 19'-9 1/2": field boards cut to 19'-0", 41 full rows, deck over fascia 19'-11 3/8" x 19'-10 1/2"

CODE	STAIRS	right stair: 16 risers — graspable handrail 34–38" on at least one side (IRC R311.7.8) IRC R311.7
CHECK	SETBACK	rear setback not entered — confirm the required rear setback and the deck-to-line distance before the frame is laid out zoning
CHECK	SETBACK	side setback not entered — confirm the required side setback and the deck-to-line distance before the frame is laid out zoning
CHECK	FOOTING	Diamond Pier: pins 50" — locate utilities (811) before driving; bearing 2,585 lb/post vs 3,300 lb allowable (manufacturer chart, confirm)
CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
CHECK	STAIRS	right stair: stringer run 14.6' — 2 carriers under the stringers at the third points, each on two 6x6 posts and footings (4.9' spans); 4 posts and footings in all IRC R311.7
INFO	LEDGER	lateral: 4 x DTT1Z (750 lb each) into house floor framing IRC R507.9.2
INFO	WIND	basic wind speed 115 mph — standard connectors
INFO	STAIRS	right stair: 16 risers @ 7.50" · 15 treads @ 11.66" · run 14'-6 7/8" · stringers 5 @ 12" OC (composite treads) IRC R311.7.5 / TimberTech
INFO	SETBACK	no address on the spec — tax rate and jurisdiction defaults used

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
5	6	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (5 blocking)
28	29	ea	2x10x20 #1 True Frame Joist SYP GC	+1 cull (19 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · 2 front rim ply · ledger)
5	6	ea	2x12x20 #1 True Frame Joist SYP GC	+1 cull (5 stair stringer)
1	2	ea	2x6x12 #1 True Frame Joist SYP GC	+1 cull (stair kicker/hanger board)
2	2	ea	4x12x20 Douglas Fir #2	exact (2 drop beam: DROP 4x12 DF · DROP 4x12 DF)
6	6	ea	2x6x12 #1 KDAT — stair carrier (2 ply) + blocking between stringers at each carrier (right, 2 carriers)	exact (2 ply x 4'-6" carrier + 4 blocks, 1 spare — per carrier)
11	11	ea	6x6x8 #2 GC	exact (yields 12 posts at ~7'-10")

Footings

NET	ORDER	UNIT	ITEM	WHY
12	12	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
12	12	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact
1	1	ea	Stair landing pad (right stair) — 4" slab, stringers bear on the pad	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
42	42	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (21 joists x 2 ends)
4	4	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	1	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	252 needed of ~350
1	1	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	628 needed of ~650
1	1	box	FastenMaster LedgerLOK 3-5/8" 50 ct	21 needed
4	4	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
46	46	ea	Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)	exact (21 joists + 2 rims at each drop beam)
8	8	ea	Simpson BC46Z post cap 6x post / 4x beam	exact
3	3	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	224 needed (bases 144 · caps 80)
4	4	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	366 needed (laminations 126 · blocking 240)
36	38	ea	7/16" x 4-1/2" HDG hex bolt + nut + 2 washers	+2 (18 rail posts x 2)
5	5	ea	Simpson LSCZ adjustable stair-stringer connector	exact (right stair, 1 per stringer at the rim)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	1	set	Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams	exact
8	8	roll	Nichigo G-Tape 3035 2" x 65' black	480' needed of 520'
2	2	roll	Nichigo G-Tape 3035 4" x 65' black	98' needed of 130'

Decking

NET	ORDER	UNIT	ITEM	WHY
41	43	ea	TimberTech Vintage Coastline 1x6x20 Square Edge	+2 (A: 41 rows @ 19'-0")
3	4	ea	TimberTech Vintage Coastline 1x6x20 Square Edge — borders	+1 (front border 19'-11 3/8" · left border 19'-10 1/2" · right border 19'-10 1/2")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
5	5	box	Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)	1726 screws (2 per board at every bearing point) · 1750 supplied
1	1	box	Cortex for TimberTech PVC, Coastline plugs — borders + dividers	110 screws

Fascia

NET	ORDER	UNIT	ITEM	WHY
6	7	ea	Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus (color TBD) (wraps the double rim, in place of the composite fascia)	+1 (left fascia 12'-0" · left fascia 7'-10" · right fascia 12'-0" · right fascia 7'-10" · front fascia 12'-0" · front fascia 7'-10 7/16")
1	1	box	Color-matched trim screws for HardieTrim, 100 ct ((color TBD))	44 color-matched screws, pre-drilled, 2 per joist line
1	1	kit	ColorPlus touch-up kit, (color TBD)	cut ends

Rail

NET	ORDER	UNIT	ITEM	WHY
8	8	ea	TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)	exact (cut to 6'-11 3/4" / 6'-3")
2	2	ea	TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt	exact
3	3	ea	TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt	exact
5	5	ea	TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt	exact
4	5	ea	Drink rail — TimberTech Vintage Coastline 1x6x16 Square Edge laid flat on the top rail, 54 LF, mitred at every turn	+1 (full-board line only, never scalloped)
8	8	kit	TimberTech drink-rail brackets, black — per 8' section (Build SOP: always TimberTech drink-rail brackets)	exact (one per bay)
1	1	ea	Rust-Oleum flat black touch-up (cut ends)	exact

Stairs

NET	ORDER	UNIT	ITEM	WHY
15	16	ea	TimberTech Vintage Coastline 1x6x12 Square Edge — treads (right stair)	+1 (15 treads x 2 boards @ 4'-0", 2 per board)
8	8	ea	TimberTech Vintage Coastline riser board 7-1/4" x 12' (right stair)	exact (16 risers @ 4'-0")
6	6	ea	TimberTech Fulton Rail 6' x 36" STAIR panel Black (right stair)	exact
8	8	ea	TimberTech Fulton 2" STAIR post Black w/ brackets, cap (right stair)	exact (top + bottom of each rail side)
1	1	ea	Fulton graspable handrail kit (right stair)	exact (4+ risers — IRC R311.7.8)

FASTENER & CONNECTOR SCHEDULE

Ledger	2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1726 field + 110 border/divider)
Rail	Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists	2x10	19'-5"	19	@ 12" OC, crown up
Picture-frame joists	2x10	19'-5"	2	centre 5 1/4" from each frame face
Side rim plies	2x10	19'-5"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Front rim plies	2x10	19'-9 7/16"	2	inner ply takes the hangers; outer ply laminated after
Ledger	2x10	19'-9 7/16"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking	2x10	10 1/2"	60	3 rows at 9'-0", 17'-7 3/4", 4'-6 3/4" from the wall
Field boards run A	Vintage Coastline	19'-0"	41	boards run parallel to the house
Stair stringers (right)	2x12	18'-8 1/8"	5	16 risers @ 7.500" · 15 treads @ 11.66" · stringers 12" OC
DROP 4x12 DF beam	4x12	19'-8 7/16"	1	y 9'-0"; posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16"
Posts under DROP 4x12 DF	6x6	7'-10"	4	field-measure each
DROP 4x12 DF beam	4x12	19'-8 7/16"	1	y 17'-7 3/4"; posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16"
Posts under DROP 4x12 DF	6x6	7'-10"	4	field-measure each
Stair carrier posts (right, carrier 1 of 2)	6x6	6'-8"	2	under the stringers — field-measure
Stair carrier (right, carrier 1 of 2)	2x6	4'-6"	2	2 ply, notched stringers bear on it; hurricane tie each stringer
Stair carrier posts (right, carrier 2 of 2)	6x6	3'-4"	2	under the stringers — field-measure
Stair carrier (right, carrier 2 of 2)	2x6	4'-6"	2	2 ply, notched stringers bear on it; hurricane tie each stringer
Left border	Vintage Coastline square	19'-10 1/2"	1	mitre at the corners; dividers on the doubled joist

MEMBER	STOCK	LENGTH	QTY	NOTE
Right border	Vintage Coastline square	19'-10 1/2"	1	mitre at the corners; dividers on the doubled joist
Front border	Vintage Coastline square	19'-11 3/8"	1	mitre at the corners; dividers on the doubled joist
Stair treads (right)	Vintage Coastline square	4'-0"	30	2 boards per tread, nosing 3/4" over the riser
Stair risers (right)	riser board	4'-0"	16	rip to 7.50" less tread thickness
Left fascia	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia	fascia	7'-10"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	7'-10"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	7'-10 7/16"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK internal

Materials (raw)	\$21,111.88
Materials × 1.15	\$24,278.66
Sales tax 4.00% on factored materials	\$971.15
Labor (rate card)	\$11,678.00
Work — carries the margin	\$36,927.81
General conditions at cost	\$1,584.00
Cost	\$38,511.81
Sell — check or ACH at 42.5% GM	\$65,806 · \$166.18/SF
Sell at the 40% floor	\$63,130
Financed	\$70,759 · \$821/mo at 6.99% × 10 yr
Gross profit	\$27,294
Engineering — outside the price, at cost	\$1,800–\$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Diamond Pier install (replaces caisson)	12	pier	\$50.00	\$600.00

LINE	QTY	UNIT	RATE	EXT
Frame	396	SF	\$10.00	\$3,960.00
Decking & fascia	396	SF	\$8.00	\$3,168.00
Railing	54	LF	\$20.00	\$1,080.00
Drink rail add-on	54	LF	\$5.00	\$270.00
Extra beam set	1	set	\$200.00	\$200.00
Stairs (right)	16	riser	\$150.00	\$2,400.00

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	\$1,584	\$4/SF x 396 SF

What the price includes — sell values

ITEM	VALUE
Foundation & structure	\$23,189
Decking & fascia	\$21,467
Railing	\$5,889
Stairs	\$11,344
Coastline drink rail on the rail	\$2,333
Site & project services	\$1,584
Total, check or ACH	\$65,806
Total, financed	\$70,759

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

20 X 20 DECK — TIMBERTECH ADVANCED PVC, FULTON + DRINK RAIL, CASTLE ROCK

2x10 SYP frame at 12", 4x12 beam, 6x6 posts — on 12 Diamond Pier footings.

TimberTech Vintage Coastline decking, Fulton rail with a Coastline drink rail.

396 SF · 1 zone · **10'-0"** above grade, engineered · **30 psf** snow design

PRICE

Total investment \$70,759 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about \$821/month

Check or ACH: \$65,806 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Foundation & structure	\$23,189
Decking & fascia	\$21,467
Railing	\$5,889
Stairs	\$11,344
Coastline drink rail on the rail	\$2,333
Site & project services	\$1,584
Total, check or ACH	\$65,806
Total, financed	\$70,759

Engineering — \$1,800 to \$3,000. Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT SELECTION

Decking	TimberTech PVC Vintage, Coastline; picture frame in Vintage Coastline. Square-shoulder boards, 3/16" gaps, matching fascia. Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Railing	TimberTech Fulton Rail, Black, 36"; Coastline drink rail on top. 54 LF.
Frame	21 2x10 #1 True Frame Joist SYP GC @ 12" OC x 19'-5"; 2-ply 2x10 front and sides; DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb).
Foundation	DP-50/50.
Design load	30 psf snow, stamped by a Colorado engineer.

ELEMENT SELECTION

Hardware Simpson ZMAX galvanized connectors. Cortex fasteners with color-matched plugs.

PLAN

19'-9 7/16" x 19'-9 1/2" outside to outside · 396 SF · 10'-0" above grade · 54 LF rail

- Stair right: 16 risers @ 7.50", 15 treads, 5 stringers, 4'-0" wide

SCHEDULE

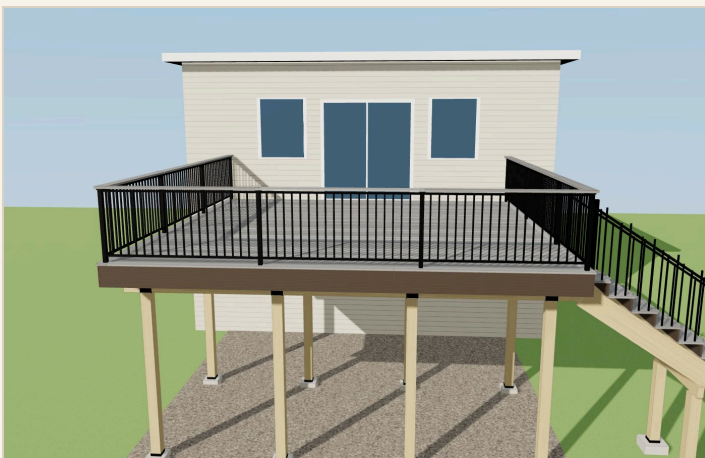
1. Engineering & ordering (2-3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811).
3. Day 2 — layout, 12 Diamond Piers driven, ledger set.
4. Days 3-5 — posts and beams set.
5. Days 6-8 — frame, flashing at every wall.
6. Days 9-10 — decking, borders first.
7. Days 11-11 — Fulton rail and drink rail.
8. Day 12 — stairs.
9. Day 13 — final clean; end of construction.

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

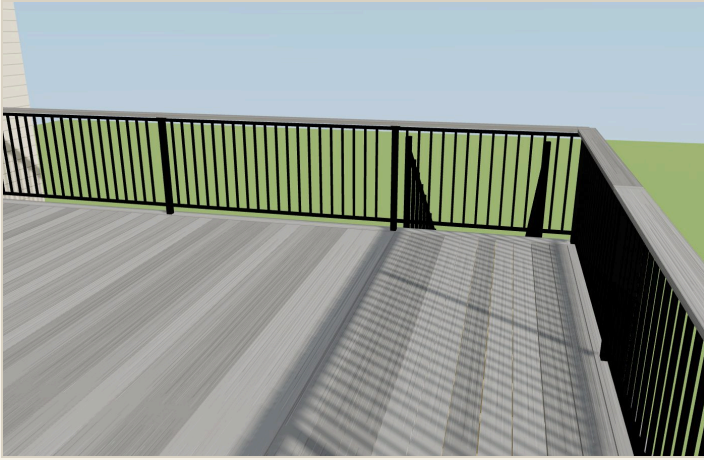
RENDERS 3D model · design intent · timbers shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings

ISOMETRICS



Isometric from the stair corner



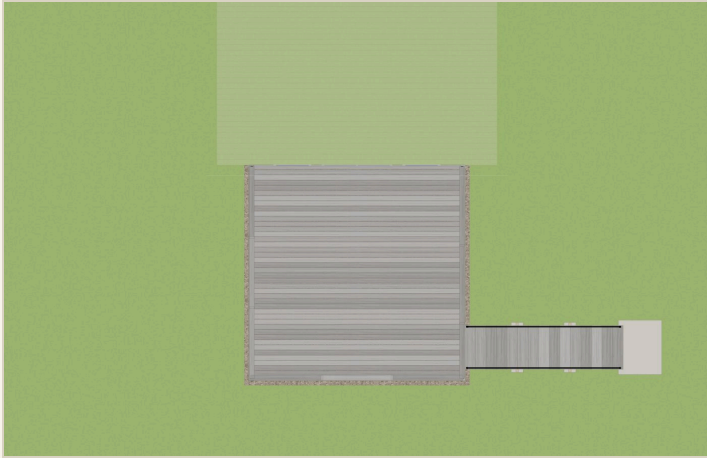
Isometric from the open corner



Low isometric

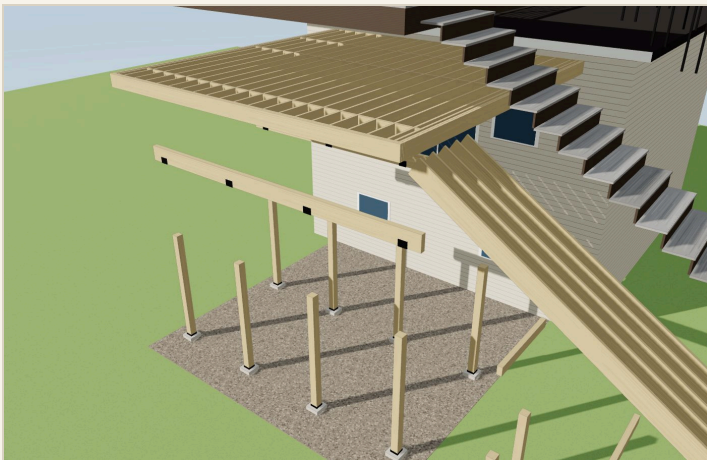


Isometric from under the frame

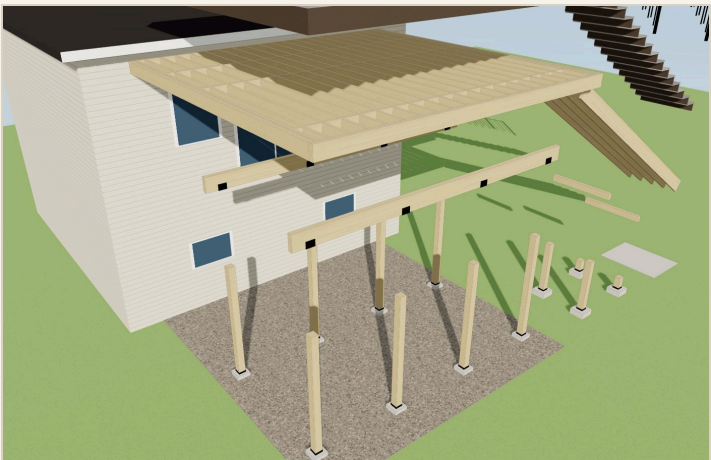


Plan from above

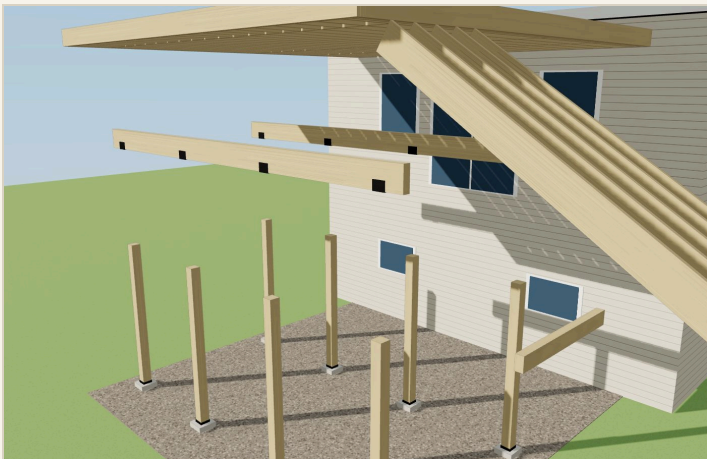
EXPLODED



Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



Exploded — low angle

G-001 General notes · design criteria

G-001

GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

GSX · Jade Pender 303-550-9558 · October 02, 2026 · Rev 0 FOR REVIEW

PROJECT

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle Rock
Castle Rock CO 80104
396 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult, Exposure C, Risk Cat. II (ASCE 7-16) — confirm special wind region
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC as adopted by Castle Rock; IRC R507 decks; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	12 x 6x6
Beams	DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 21 2x10 #1 True Frame Joist SYP GC @ 12" OC x 19'-5"
Joists	2-ply 2x10 front and sides
Rims	2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
Ledger	TimberTech Vintage Coastline — 41 rows parallel to the house, picture frame in Vintage Coastline, 3/16" gaps, Cortex
Decking	Fulton 36" Black: 8 bays, 10 posts, 54 LF + drink rail
Rail	

SHEET INDEX

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A-001	Isometric · renders
S-100	Foundation plan
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A-301	Section + front elevation
D-501	Details
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

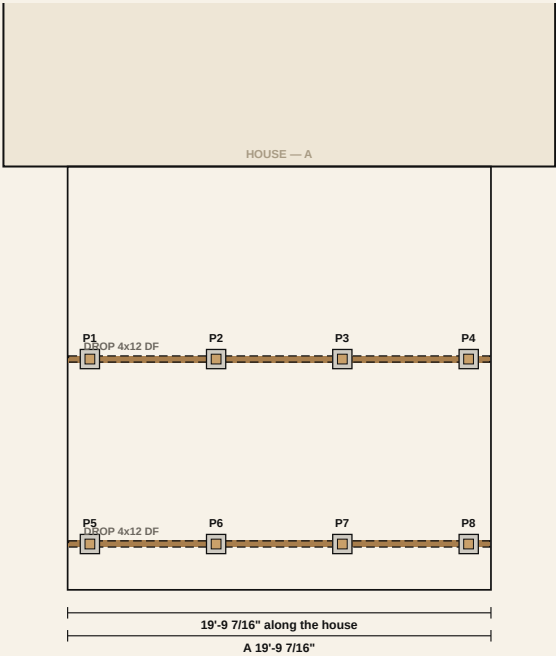
Setbacks not on file — verify the plat / ILC before layout

FLAGS (INTERNAL — remove before the permit submittal)

CODE	right stair: 16 risers — graspable handrail 34–38" on at least...
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No scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

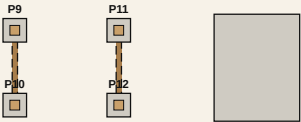


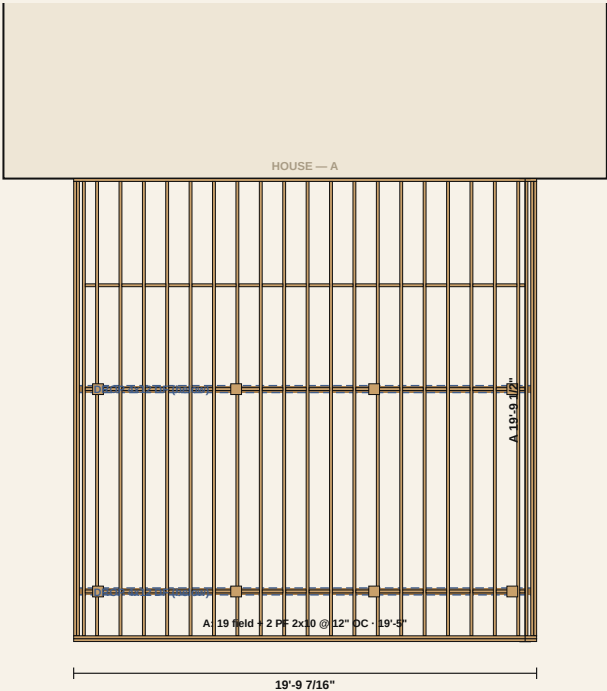
FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins),
3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	1'-0 1/2"	9'-0"	A	2,585 lb
P2	6'-11 5/16"	9'-0"	A	2,585 lb
P3	12'-10 1/8"	9'-0"	A	2,585 lb
P4	18'-8 15/16"	9'-0"	A	2,585 lb
P5	1'-0 1/2"	17'-7 3/4"	A	2,585 lb
P6	6'-11 5/16"	17'-7 3/4"	A	2,585 lb
P7	12'-10 1/8"	17'-7 3/4"	A	2,585 lb
P8	18'-8 15/16"	17'-7 3/4"	A	2,585 lb
P9	24'-7 3/4"	15'-0 1/2"	stair	carrier
P10	24'-7 3/4"	18'-6 1/2"	stair	carrier
P11	29'-6"	15'-0 1/2"	stair	carrier
P12	29'-6"	18'-6 1/2"	stair	carrier

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,585 lb

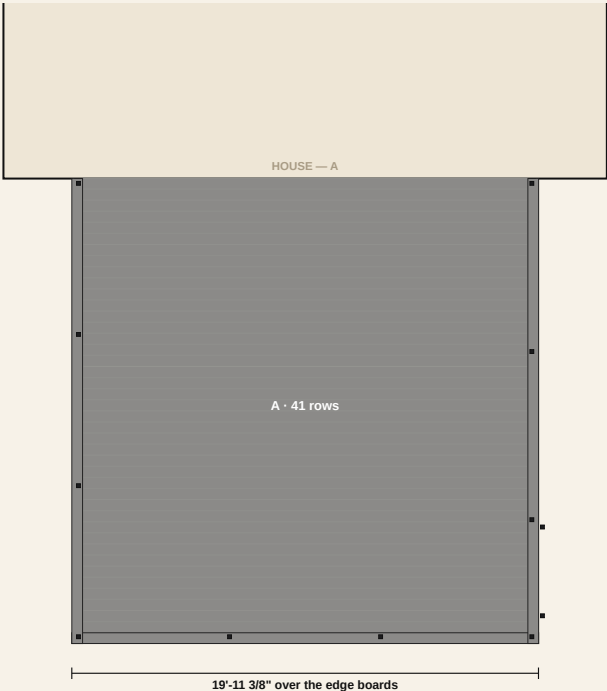




FRAMING NOTES

- 21 2x10 #1 True Frame Joist SYP GC @ 12"
- OC x 19'-5"
- 2-ply 2x10 front and sides
- 2x10 ledger x 19'-9 7/16" total ·
- LedgerLOK 2 rows 2" from top/bottom,
- staggered 12" OC, first 3" from each end
- blocking one row over the beam,
- staggered
- G-Tape 2" every joist, PF joist,
- blocking and the ledger top · 4" on both
- rims and the beam(s)
- DROP 4x12 DF: 19'-8 7/16", 4 posts at
- 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" /
- 18'-8 15/16" (worst 2,585 lb); DROP 4x12
- Connectors:
- Simpson LUS28Z face-mount hanger
- 2x8/2x10: 42
- Simpson HUCQ210-2-SDS concealed-flange
- double 2x10 hanger (SDS screws
- included): 4
- Simpson DTT12Z lateral tie kit w/ SDS
- screws: 4
- Simpson H2.5AZ hurricane tie (joists to
- the drop beam, hidden): 46
- Simpson BC46Z post cap 6x post / 4x
- beam: 8
- 7/16" x 4-1/2" HDG hex bolt + nut + 2
- washers: 36
- Simpson LSCZ adjustable stair-stringer
- connector: 5

A-101 Decking plan



DECKING

Field: TimberTech Vintage Coastline, 5.5" x 1", 3/16" gaps

Border: Vintage Coastline

Boards run parallel to the house; edge overhang 1" over the fascia

Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1726 field + 110 border/divider)

Rows per zone: A 41 (no rips)

A-201 Railing plan + elevation

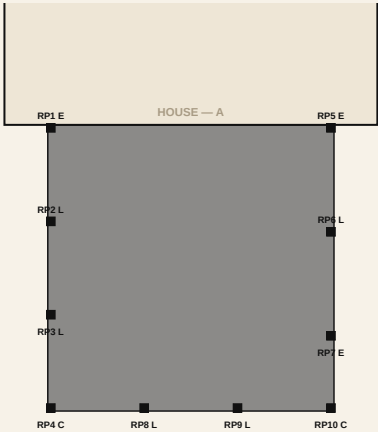
A-201

RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

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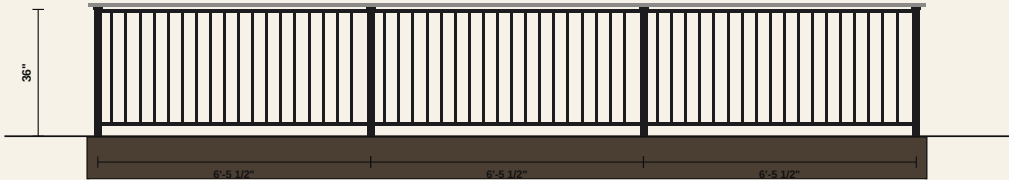
RAILING

Fulton 36" Black: 8 bays, 10 posts, 54 LF + drink rail

Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

Bays: left 6'-5 1/2" · left 6'-5 1/2" · left 6'-5 1/2" · right 7'-2 1/4" · right 7'-2 1/4" · front 6'-5 1/2" · front 6'-5 1/2" · front 6'-5 1/2"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

A-301

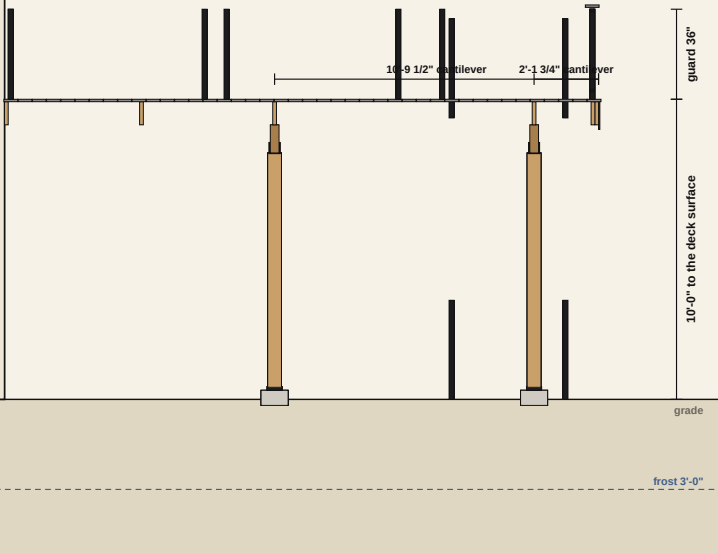
SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

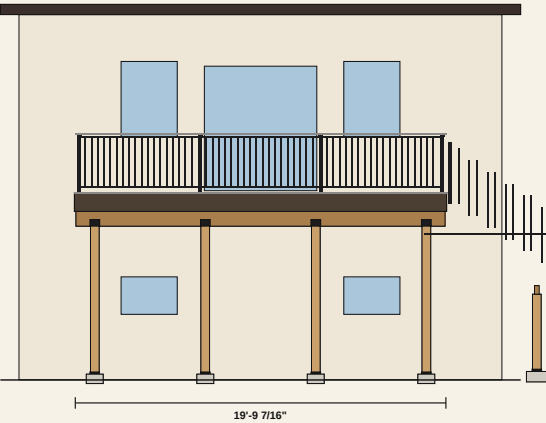
20 x 20 Deck — TimberTech Advanced PVC, Fulton + drink rail, Castle...

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SECTION through zone A at 9'-10 3/4" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb); DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8"

Posts 12 x 6x6 on DP-50/50

2x10 ledger x 19'-9 7/16" total - LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

1 LEDGER AT THE HOUSE

2x10 ledger x 19'-9 7/16" total - LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end - 21 fasteners

2 POST · BEAM · CAP

DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16" (worst 2,585 lb). DROP 4x12 DF: 19'-8 7/16", 4 posts at 1'-

3 FOOTING · POST BASE

4 x 50" pins · DP-50/50

4 RAIL POST · GUARD

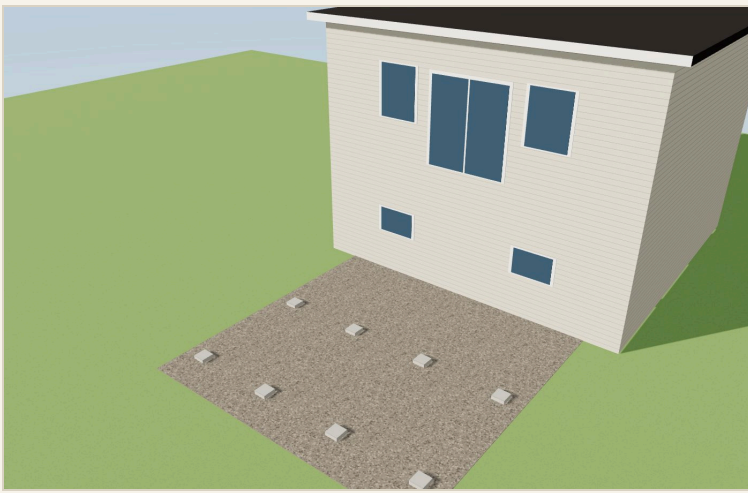
Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn

5 DECK EDGE · BORDER

Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1726 field + 110 border/divider)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.



Step 1 • Footings — 12 Diamond Pier DP-50/50 driven-pin footings

T-400 • Step 1 • Footings

GOES IN

- (12) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (12) Simpson ABA66Z adjustable post base 6x6 on the pier bolt
- (1) Stair landing pad (right stair) — 4" slab, stringers bear on the pad

HOW

- Call 811 and locate before any hole or pin.
- Pull the beam lines from the house wall:
DROP 4x12 DF at 9'-0" from the reference wall, posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16"; DROP 4x12 DF at 17'-7 3/4" from the reference wall, posts at 1'-0 1/2" / 6'-11 5/16" / 12'-10 1/8" / 18'-8 15/16".
- Set 12 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 • Posts — 6x6 #2 GC on ZMAX bases

T-401 • Step 2 • Posts

GOES IN

- (11) 6x6x8 #2 GC
- (12) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): DROP 4x12 DF posts 7'-10" / DROP 4x12 DF posts 7'-10" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff

T-403 · Step 4 · Beams

GOES IN

- (1) 2x6x12 #1 True Frame Joist SYP GC
- (2) 4x12x20 Douglas Fir #2
- (6) 2x6x12 #1 KDAT — stair carrier (2 ply) + blocking between stringers at each carrier (right, 2 carriers)
- (11) 6x6x8 #2 GC
- (8) Simpson BC46Z post cap 6x post / 4x beam

HOW

- DROP 4x12 DF: 19'-8 7/16" one piece, 4 posts @ 5'-10 13/16" max
- DROP 4x12 DF: 19'-8 7/16" one piece, 4 posts @ 5'-10 13/16" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened



Step 4 · Beams — DROP 4x12 DF (4 posts) · DROP 4x12 DF (4 posts)

T-404 · Step 5 · Frame

GOES IN

- (5) 2x10x12 #1 True Frame Joist SYP GC
- (28) 2x10x20 #1 True Frame Joist SYP GC
- (42) Simpson LUS28Z face-mount hanger 2x8/2x10
- (4) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (46) Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden)
- (1) Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams
- (8) Nichigo G-Tape 3035 2" x 65' black

HOW

- 2x10 ledger x 19'-9 7/16" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 21 fasteners
- A: 19 joists @ 12" OC x 19'-5". Crown up, hangers at the ledger.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 5 · Frame — 2x10 joists at 12", rims, ledger, blocking over the beam

T-405 · Step 6 · Decking

GOES IN

- (41) TimberTech Vintage Coastline 1x6x20 Square Edge
- (3) TimberTech Vintage Coastline 1x6x20 Square Edge — borders
- (6) Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus (color TBD) (wraps the double rim, in place of the composite fascia)
- (1) Color-matched trim screws for HardieTrim, 100 ct ((color TBD))
- (1) ColorPlus touch-up kit, (color TBD)
- (5) Cortex for TimberTech PVC, Coastline plugs (350 ct / 100 SF)
- (1) Cortex for TimberTech PVC, Coastline plugs — borders + dividers

HOW

- Cortex face screws with color-matched plugs: 2 per board at every joist and at both ends of every run, color-matched plugs (1726 field + 110 border/divider)
- Borders first, then the field.
- Rows per zone: A 41. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 6 · Decking — TimberTech Vintage Coastline, Coastline border, hidden fasteners

T-406 · Step 7 · Rail

GOES IN

- (8) TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)
- (2) TimberTech Fulton 2" CORNER post 36" Black w/ brackets, cap, skirt
- (3) TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt
- (5) TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt
- (4) Drink rail — TimberTech Vintage Coastline 1x6x16 Square Edge laid flat on the top rail, 54 LF, mitred at every turn
- (8) TimberTech drink-rail brackets, black — per 8' section (Build SOP: always TimberTech drink-rail brackets)
- (1) Rust-Oleum flat black touch-up (cut ends)

HOW

- Fulton 36" — 10 posts (2 CORNER, 3 END, 5 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post; drink rail board on TimberTech drink-rail brackets, mitred at every turn
- Drink rail on TimberTech drink-rail brackets, mitred at every turn, plugs color-matched.

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction



Step 7 · Rail — Fulton 36" Black, drink rail on top



The interactive 3D model lives at the online version of this document.

GS Exterior Experts · Deck Division

Prices: rate card · GC at cost · 42.5% GM · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.